



Lab No 1

Cloud Computing

**Lab Title: GitHub Identity, Student Portal Registration,
and Ubuntu Server Installation**

Submitted To: Engr. Waqas Saleem

Submitted By: Laiba Saleem

Registration No: 2024-BSE-066

Course Code: SE-665

Section: B

GitHub Username: LaibaSaleem066

**DEPARTMENT OF SOFTWARE ENGINEERING
SESSION (2024-2028)**

Task 1

Create and Configure Your GitHub Account

Description:

Set up a GitHub account under the username LaibaSaleem066 and updated the public display name to Laiba Saleem within the account settings.

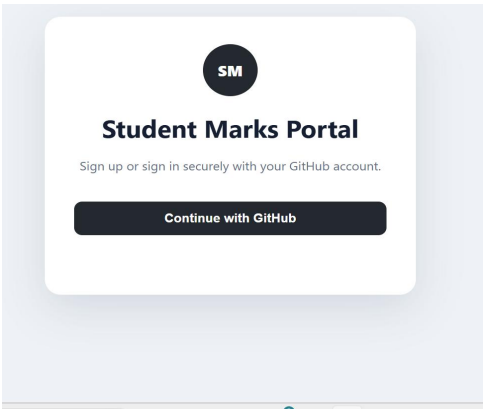
The screenshot shows the GitHub profile page for user LaibaSaleem066. The header includes the username, a search bar, and navigation links for Overview, Repositories, Projects, Packages, and Stars. The profile section features a circular profile picture with a teal and white geometric pattern, the name 'Laiba Saleem', the username 'LaibaSaleem066', and an 'Edit profile' button. To the right, it states 'You don't have any public repositories yet.' Below this is a '1 contribution in the last year' section with a calendar grid showing contributions from September to August. A 'Contribution settings' dropdown is visible. At the bottom, a message explains the contribution graph. The settings section on the left lists 'Public profile' as the active tab, with other options like Account, Appearance, Accessibility, Notifications, Access, Billing and licensing, Emails, and Password and authentication. The 'Public profile' settings include fields for Name (Laiba Saleem), Public email (a dropdown to select a verified email), and Bio. A profile picture upload section is also present.

Task 2

Register in the Student Marks Portal

Description:

Logged into the Student Marks Portal using GitHub authentication and enrolled in Section B of Cloud Computing (SE-665) with Registration No 2024-BSE-66.



STUDENT MARKS PORTAL

Academic Record

Sign out



LaibaSaleem066

GitHub authenticated

Registration

2024-BSE-66

GitHub username

[LaibaSaleem066](#)

Email

fa24b1-se-066@fjwu.edu.pk

Approved courses

1

+ Enroll in another course

APPROVED COURSES

Course Marks

Published

Cloud Computing (SE-665) Section B

Class Participation

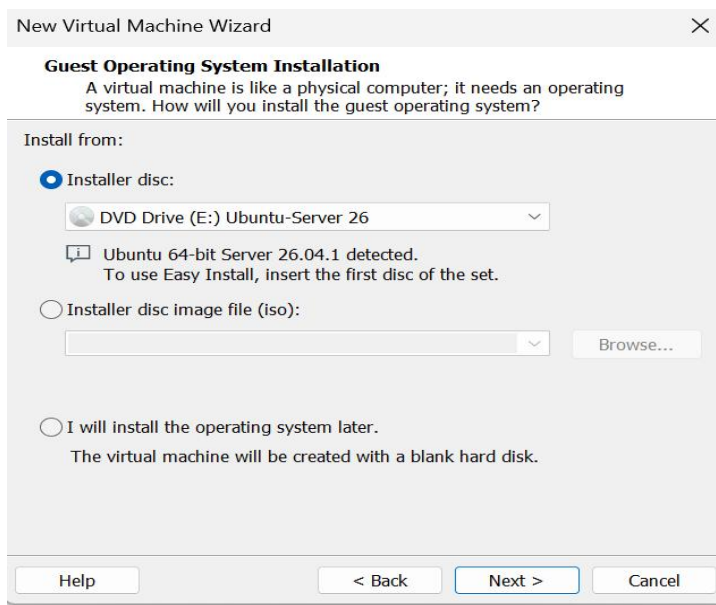
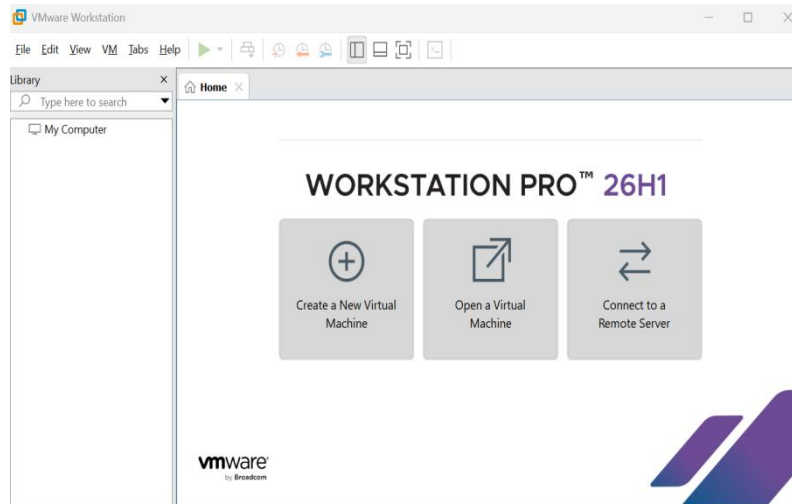
ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
Class Participation	0	10	0.0%
Total	0	10	0.0%

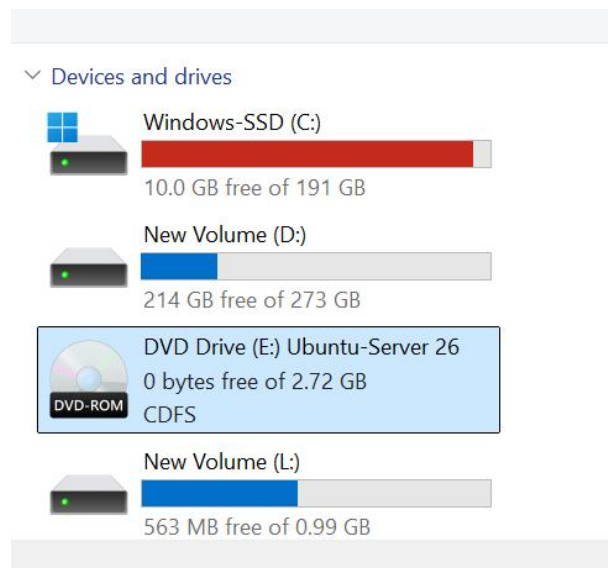
Task 3

Download VMware Workstation Pro and Ubuntu Server

Description:

Installed VMware Workstation Pro on the Windows host machine, downloaded the latest Ubuntu Server LTS ISO image, and verified that at least 20 GB of free storage space was available for the virtual machine.



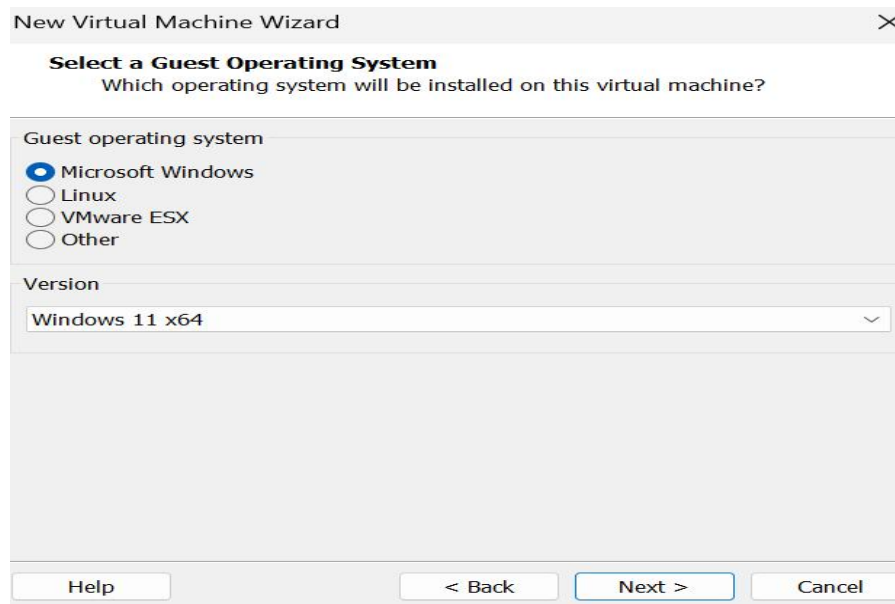


Task 4

Create the Ubuntu Server Virtual Machine

Description:

Set up a new Virtual Machine in VMware Workstation Pro using the Typical configuration setup, mounted the downloaded Ubuntu Server ISO, allocated 20 GB of disk space, and started the VM installation process.



New Virtual Machine Wizard

Easy Install Information
This is used to install Ubuntu 64-bit.

Personalize Linux

Full name:

User name:

Password:

Confirm:

Help < Back Next > Cancel

```

Starting netplan-configure.service - Netplan Backend Configuration...
Starting plymouth-read-write.service - Plymouth To Write Out Runtime Data...
Starting snapd.apparmor.service - Profiles managed internally by snapd...
Starting systemd-binfmt.service - Set Up Additional Binary Formats...
Starting systemd-tmpfiles-setup.service - Create System Files and Directories...
Starting ufw.service - Uncomplicated firewall...
[ OK ] Finished finalrd.service - Create runtime dir for shutdown pivot root.
[ OK ] Finished plymouth-read-write.service - Plymouth To Write Out Runtime Data.
[ OK ] Finished ufw.service - Uncomplicated firewall.
[ OK ] Finished systemd-tmpfiles-setup.service - Create System Files and Directories.
Starting ldconfig.service - Rebuild Dynamic Linker Cache...
Starting systemd-journal-catalog-update.service - Rebuild Journal Catalog...
[ OK ] Started vgauth.service - Authenticate virtual machines hosted on VMware.
[ OK ] Started open-vm-tools.service - Set up virtual machines hosted on VMware.
[ OK ] Finished systemd-udev-trigger.service - Coldplug All udev Devices.
[ OK ] Started systemd-ask-password-console.service - Requests to Console Directory Watch.
[ OK ] Reached target cryptsetup.target - Local Encrypted Volumes.
[ OK ] Finished systemd-journal-catalog-update.service - Rebuild Journal Catalog.
[ OK ] Finished netplan-configure.service - Netplan Backend Configuration.
[ OK ] Finished kdump-tools.service - Kernel crash dump capture service.
Mounting proc-sys-fs-binfmt_misc.mount - Executable File Formats File System...
[ OK ] Mounted proc-sys-fs-binfmt_misc.mount - Executable File Formats File System.
[ OK ] Finished systemd-binfmt.service - Set Up Additional Binary Formats.
[ 22.310223] piix4_smbus 0000:00:07.3: SMBus Host Controller not enabled!

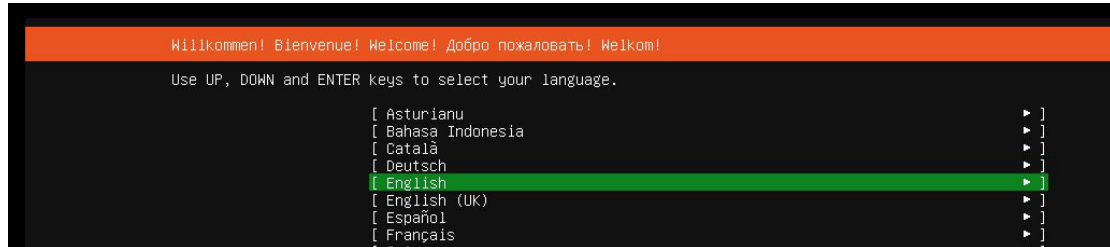
```

Task 5

Choose the Installation Language

Description:

Picked English as the default installation language in the text-based Ubuntu Server setup wizard to continue configuring the OS.



Task 6

Configure the Keyboard Layout

Description:

Selected the base keyboard layout and matching input variant to ensure correct key mapping during system usage.



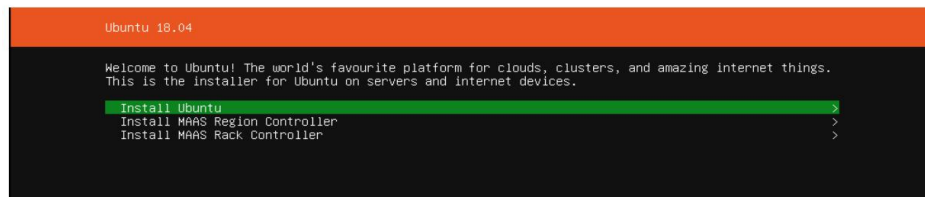
Task 7

Choose the Ubuntu Server Installation Type

Description:

Opted for the standard Ubuntu Server base installation, skipping extra provisioning features like MAAS or automated enterprise tools.

Instructional reference

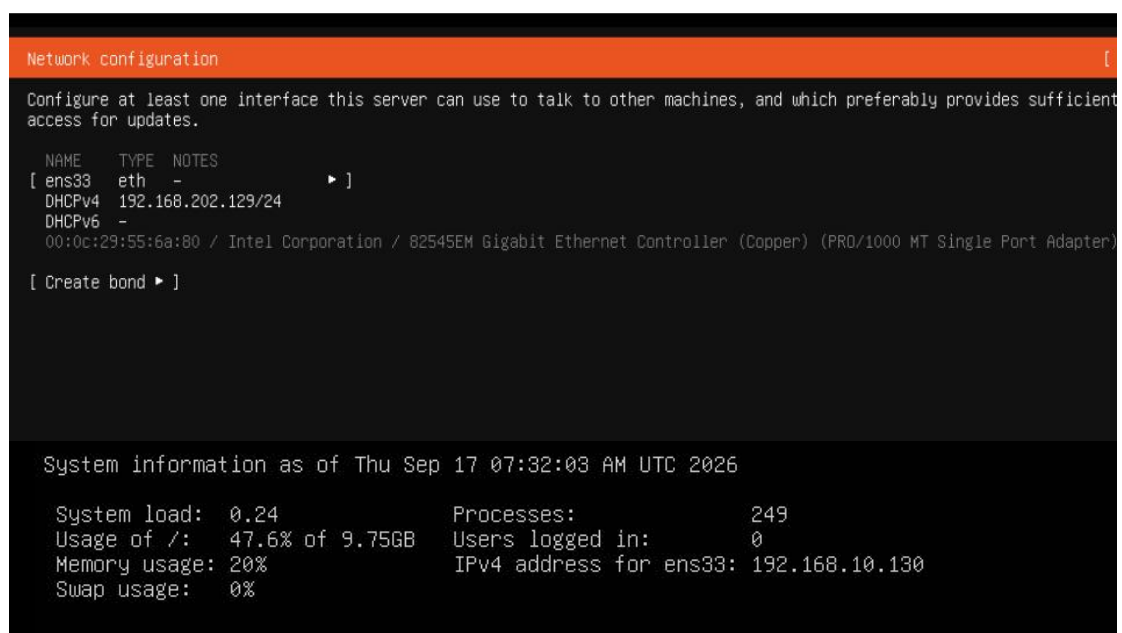


Task 8

Configure the Network

Description:

Ensured the installer detected the network adapter and confirmed it automatically received an IPv4 address via DHCP.



Task 9

Configure Storage

Description:

Selected the guided storage option to automatically partition the VMware virtual disk, then reviewed the resulting storage layout

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY
MOUNT POINT  SIZE  TYPE  DEVICE TYPE
[ /           10.000G  new ext4  new LVM logical volume  ▶ ]
[ /boot       1.771G  new ext4  new partition of local disk ▶ ]

AVAILABLE DEVICES
DEVICE                TYPE                SIZE
[ ubuntu-vg (new)      LVM volume group    18.222G  ▶ ]
free space            8.222G  ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES
DEVICE                TYPE                SIZE
[ ubuntu-vg (new)      LVM volume group    18.222G  ▶ ]
ubuntu-lv             new, to be formatted as ext4, mounted at / 10.000G  ▶ ]

[ /dev/sda             local disk           20.000G  ▶ ]
partition 1            new, BIOS grub spacer 1.000M  ▶ ]
partition 2            new, to be formatted as ext4, mounted at /boot 1.771G  ▶ ]
partition 3            new, PV of LVM volume group ubuntu-vg 18.225G  ▶ ]

[ Done ]
[ Reset ]
[ Back ]
```

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY
MOUNT POINT  SIZE  TYPE  DEVICE TYPE
[ /           10.000G  new ext4  new LVM logical volume  ▶ ]
[ /boot       1.771G  new ext4  new partition of local disk ▶ ]

AVAILABLE DEVICES
DEVICE                TYPE                SIZE
[ ubuntu-vg (new)      LVM volume group    18.222G  ▶ ]
free space            8.222G  ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]
```

Task 10

Select the Installation Disk

Description:

Chose the target VMware virtual drive for installation and verified its total capacity and details before proceeding with formatting.

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY

MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /             10.000G new ext4 new LVM logical volume ▶ ]
[ /boot         1.771G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
free space      8.222G ▶

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
ubuntu-lv       new, to be formatted as ext4, mounted at / 10.000G ▶

[ /dev/sda      local disk 20.000G ▶ ]
partition 1     new, BIOS grub spacer 1.000M ▶
partition 2     new, to be formatted as ext4, mounted at /boot 1.771G ▶
partition 3     new, PV of LVM volume group ubuntu-vg 18.225G ▶

[ Done ]
[ Reset ]
[ Back ]
```

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY

MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /             10.000G new ext4 new LVM logical volume ▶ ]
[ /boot         1.771G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
free space      8.222G ▶

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
ubuntu-lv       new, to be formatted as ext4, mounted at / 10.000G ▶

[ /dev/sda      local disk 20.000G ▶ ]
partition 1     new, BIOS grub spacer 1.000M ▶
partition 2     new, to be formatted as ext4, mounted at /boot 1.771G ▶
partition 3     new, PV of LVM volume group ubuntu-vg 18.225G ▶

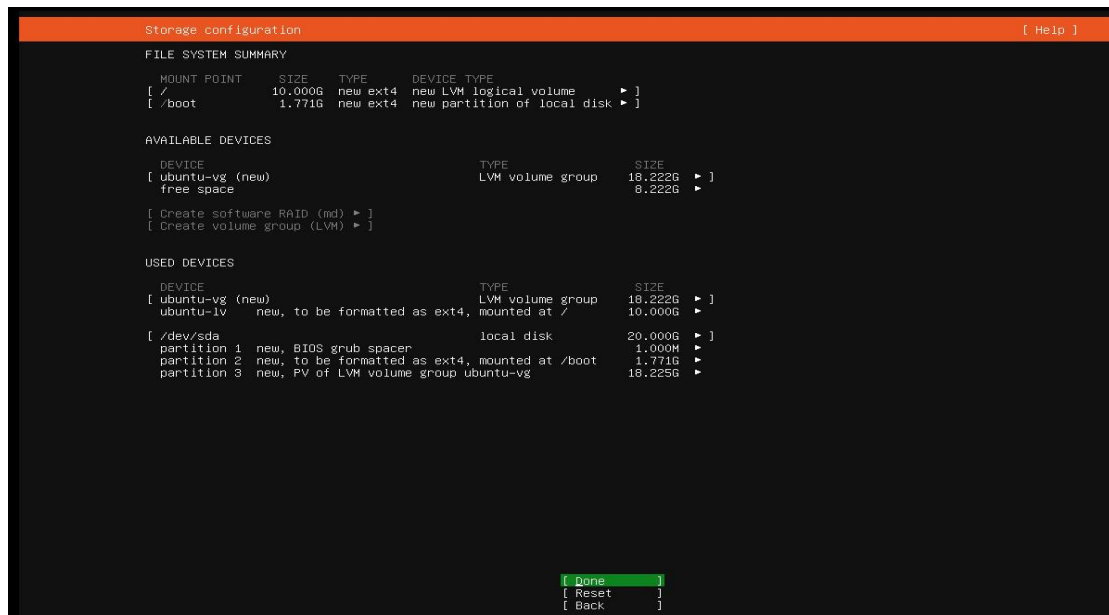
[ Done ]
[ Reset ]
[ Back ]
```

Task 11

Review the Partitions

Description:

Examined the generated file system layout and mount points created by the installer to confirm proper drive partitioning on the virtual machine



Task 12

Confirm the Storage Changes

Description:

Accepted the warning prompt regarding drive formatting and confirmed the partition scheme to begin writing changes to the virtual disk.

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY

MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /             10.000G new ext4 new LVM logical volume ▶ ]
[ /boot         1.771G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
free space      8.222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
ubuntu-lv       new, to be formatted as ext4, mounted at / 10.000G ▶ ]

[ /dev/sda      local disk 20.000G ▶ ]
partition 1     new, BIOS grub spacer 1.000M ▶ ]
partition 2     new, to be formatted as ext4, mounted at /boot 1.771G ▶ ]
partition 3     new, PV of LVM volume group ubuntu-vg 18.225G ▶ ]

[ Done ]
[ Reset ]
[ Back ]
```

Task 13

Configure the Ubuntu Profile

Description:

Entered the primary administrator credentials and server hostname in the installation wizard, matching the actual student identity and assigned GitHub handle.

```
Profile configuration [ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Laiba Saleem
```

```
Your servers name: ubuntu
The name it uses when it talks to other computers.
```

```
Pick a username: laibasaleem066

Profile configuration [ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Laiba Saleem

Your servers name: ubuntu
The name it uses when it talks to other computers.

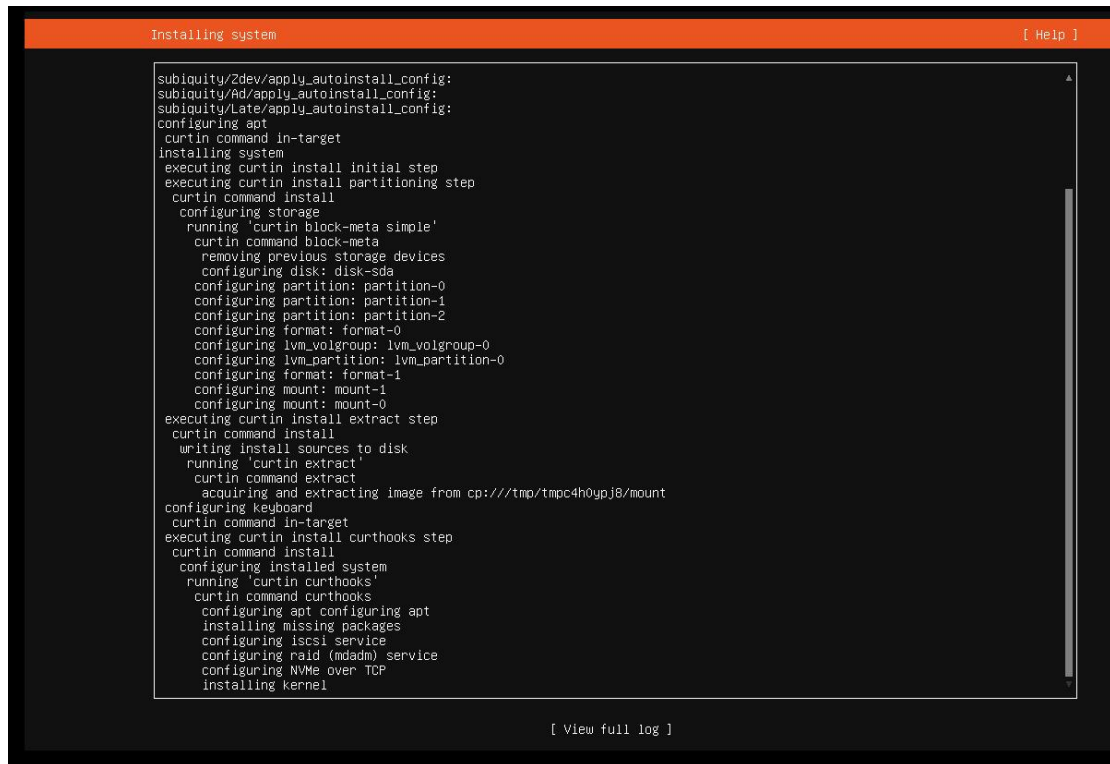
Pick a username: laibasaleem066
```

Task 14

Complete the Software Installation

Description:

Started the base OS package extraction and setup process, letting it run without interruption until the core system installation was finished.



```
Installing system [ Help ]
subiquity/2dev/apply_autoinstall_config:
subiquity/4d/apply_autoinstall_config:
subiquity/late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
curtin command install
configuring storage
running 'curtin block-meta simple'
curtin command block-meta
removing previous storage devices
configuring disk: disk-sda
configuring partition: partition-0
configuring partition: partition-1
configuring partition: partition-2
configuring format: format-0
configuring lvm_volgroup: lvm_volgroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpc4h0ypj8/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
[ View full log ]
```

Task 15

Reboot and Verify the Ubuntu Identity

Description:

Restarted the virtual machine after installation and logged into the command-line terminal to verify the custom server hostname and shell prompt settings.

Installation complete!

[Help]

```
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpc4h0ypj8/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/Late/run_builtins:
subiquity/Late/run_user_supplied:
```

[View full log]

[Reboot Now]

```
[ 24.770599] cloud-init[1171]: The key's randomart image is:
[ 24.770662] cloud-init[1171]: +---[ECDSA 256]---+
[ 24.770690] cloud-init[1171]: |                    |
[ 24.770724] cloud-init[1171]: |                    |
[ 24.770748] cloud-init[1171]: |                    |
[ 24.770770] cloud-init[1171]: |                    |
[ 24.773264] cloud-init[1171]: | S oo+ B.= |
[ 24.773681] cloud-init[1171]: | .+o+.@o |
[ 24.773946] cloud-init[1171]: | = o*== |
[ 24.774125] cloud-init[1171]: | +.*oX+. |
[ 24.774259] cloud-init[1171]: | .+O+B++E |
[ 24.774442] cloud-init[1171]: +----[SHA256]-----+
[ 24.774697] cloud-init[1171]: Generating public/private ed25519 key pair.
[ 24.777953] cloud-init[1171]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 24.780098] cloud-init[1171]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 24.782195] cloud-init[1171]: The key fingerprint is:
[ 24.784347] cloud-init[1171]: SHA256:/24Mh1x27xMI74IWqHpYKCdtSnqkWUphmrPw7vC7nfk root@ubuntu
[ 24.784988] cloud-init[1171]: The key's randomart image is:
[ 24.787380] cloud-init[1171]: +---[ED25519 256]---+
[ 24.790242] cloud-init[1171]: |                    |
[ 24.793243] cloud-init[1171]: |                    |
[ 24.793861] cloud-init[1171]: |                    |
[ 24.794362] cloud-init[1171]: | o . . o o o |
[ 24.794662] cloud-init[1171]: | oo.. . .S= . . . |
[ 24.794847] cloud-init[1171]: | BoB o =.+ . o |
[ 24.795033] cloud-init[1171]: | */ + . .+ . |
[ 24.795171] cloud-init[1171]: | Bo=. .o .o |
[ 24.795297] cloud-init[1171]: | +*+.E oo |
[ 24.795424] cloud-init[1171]: +----[SHA256]-----+

ubuntu login: ^^_
```

```
ubuntu login: laibasaleem066
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 04:55:09 AM UTC 2026

System load:  0.0           Processes:      221
Usage of /:   50.7% of 9.75GB Users logged in: 0
Memory usage: 22%          IPv4 address for ens33: 192.168.10.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

laibasaleem066@ubuntu:~$ _
```

```
ubuntu login: laibasaleem066
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 04:55:09 AM UTC 2026

System load:  0.0           Processes:      221
Usage of /:   50.7% of 9.75GB Users logged in: 0
Memory usage: 22%          IPv4 address for ens33: 192.168.10.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

laibasaleem066@ubuntu:~$ _
```

Task 16

Find the Ubuntu Server IP Address

Description:

Ran network diagnostic commands in the terminal to discover the dynamically allocated IPv4 address on the active virtual network adapter.

```
laibasaleem066@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
```

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:fb:ab:7f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29fbab7f
    inet 192.168.10.130/24 metric 100 brd 192.168.10.255 scope global dynamic ens33
        valid_lft 1596sec preferred_lft 1596sec
    inet6 fe80::20c:29ff:fe7b:ab7f/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
laibasaleem066@ubuntu:~$
```

Task 17

Connect to Ubuntu Server from Windows

Description:

Initiated an SSH session from the Windows host using PowerShell/Command Prompt to connect to the VM's IP, accepted the security host key fingerprint, and successfully logged in.

```
C:\Users\User>ssh laibasaleem066@192.168.10.130
```

```
C:\Users\User>ssh laibasaleem066@192.168.10.130
The authenticity of host '192.168.10.130 (192.168.10.130)' can't be established.
ED25519 key fingerprint is SHA256:/24MhlxZ7xMI74IWqHpYKCdtSnqkWUphmrPw7vC7nfk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 06:32:51 AM UTC 2026

System load:  0.08                Processes:            223
Usage of /:   47.6% of 9.75GB     Users logged in:     0
Memory usage: 25%                IPv4 address for ens33: 192.168.10.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

laibasaleem066@ubuntu:~$ |
```

Task 18

Prepare Lab1 Solution PDF

Description:

Assembled all completed task summaries, image captions, and verification screenshots into a clean, formatted report featuring an official cover page, then saved the final document as a PDF in the designated submission directory.

Lab Number & Title: Lab 01 - Ubuntu Server Installation

Student Name: Laiba Saleem

Registration Number: 2024-BSE-66

Course Code: SE-665

Section: Section B

GitHub Username: LaibaSaleem066



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